

Raphson as in `rtsafe`. Sometimes you are *only* interested in real roots, in which case the strategy is complete. Otherwise, you then go after quadratic factors of the form (9.5.1) by any of a variety of methods. One such is Bairstow's method, which we will discuss below in the context of root polishing. Another is Muller's method, which we here briefly discuss.

### 9.5.2 Muller's Method

*Muller's method* generalizes the secant method but uses quadratic interpolation among three points instead of linear interpolation between two. Solving for the zeros of the quadratic allows the method to find complex pairs of roots. Given *three* previous guesses for the root  $x_{i-2}$ ,  $x_{i-1}$ ,  $x_i$ , and the values of the polynomial  $P(x)$  at those points, the next approximation  $x_{i+1}$  is produced by the following formulas,

$$\begin{aligned} q &\equiv \frac{x_i - x_{i-1}}{x_{i-1} - x_{i-2}} \\ A &\equiv qP(x_i) - q(1 + q)P(x_{i-1}) + q^2P(x_{i-2}) \\ B &\equiv (2q + 1)P(x_i) - (1 + q)^2P(x_{i-1}) + q^2P(x_{i-2}) \\ C &\equiv (1 + q)P(x_i) \end{aligned} \tag{9.5.2}$$

followed by

$$x_{i+1} = x_i - (x_i - x_{i-1}) \frac{2C}{B \pm \sqrt{B^2 - 4AC}} \tag{9.5.3}$$

where the sign in the denominator is chosen to make its absolute value or modulus as large as possible. You can start the iterations with any three values of  $x$  that you like, e.g., three equally spaced values on the real axis. Note that you must allow for the possibility of a complex denominator, and subsequent complex arithmetic, in implementing the method.

Muller's method is sometimes also used for finding complex zeros of analytic functions (not just polynomials) in the complex plane, for example in the IMSL routine `ZANLY` [3].

### 9.5.3 Laguerre's Method

The second school regarding overall strategy happens to be the one to which we belong. That school advises you to use one of a very small number of methods that will converge (though with greater or lesser efficiency) to all types of roots: real, complex, single, or multiple. Use such a method to get tentative values for all  $n$  roots of your  $n$ th degree polynomial. Then go back and polish them as you desire.

*Laguerre's method* is by far the most straightforward of these general, complex methods. It does require complex arithmetic, even while converging to real roots; however, for polynomials with all real roots, it is guaranteed to converge to a root from any starting point. For polynomials with some complex roots, little is theoretically proved about the method's convergence. Much empirical experience, however, suggests that nonconvergence is extremely unusual and, further, can almost always be fixed by a simple scheme to break a nonconverging limit cycle. (This is implemented in our routine below.) An example of a polynomial that requires this cycle-breaking scheme is one of high degree ( $\gtrsim 20$ ), with all its roots just outside of

the complex unit circle, approximately equally spaced around it. When the method converges on a simple complex zero, it is known that its convergence is third order.

In some instances the complex arithmetic in the Laguerre method is no disadvantage, since the polynomial itself may have complex coefficients.

To motivate (although not rigorously derive) the Laguerre formulas we can note the following relations between the polynomial and its roots and derivatives:

$$P_n(x) = (x - x_0)(x - x_1) \dots (x - x_{n-1}) \quad (9.5.4)$$

$$\ln |P_n(x)| = \ln |x - x_0| + \ln |x - x_1| + \dots + \ln |x - x_{n-1}| \quad (9.5.5)$$

$$\frac{d \ln |P_n(x)|}{dx} = +\frac{1}{x - x_0} + \frac{1}{x - x_1} + \dots + \frac{1}{x - x_{n-1}} = \frac{P'_n}{P_n} \equiv G \quad (9.5.6)$$

$$\begin{aligned} -\frac{d^2 \ln |P_n(x)|}{dx^2} &= +\frac{1}{(x - x_0)^2} + \frac{1}{(x - x_1)^2} + \dots + \frac{1}{(x - x_{n-1})^2} \\ &= \left[ \frac{P'_n}{P_n} \right]^2 - \frac{P''_n}{P_n} \equiv H \end{aligned} \quad (9.5.7)$$

Starting from these relations, the Laguerre formulas make what Acton [1] nicely calls “a rather drastic set of assumptions”: The root  $x_0$  that we seek is assumed to be located some distance  $a$  from our current guess  $x$ , while *all other roots* are assumed to be located at a distance  $b$ ,

$$x - x_0 = a, \quad x - x_i = b, \quad i = 1, 2, \dots, n - 1 \quad (9.5.8)$$

Then we can express (9.5.6) and (9.5.7) as

$$\frac{1}{a} + \frac{n - 1}{b} = G \quad (9.5.9)$$

$$\frac{1}{a^2} + \frac{n - 1}{b^2} = H \quad (9.5.10)$$

which yields as the solution for  $a$

$$a = \frac{n}{G \pm \sqrt{(n - 1)(nH - G^2)}} \quad (9.5.11)$$

where the sign should be taken to yield the largest magnitude for the denominator. Since the factor inside the square root can be negative,  $a$  can be complex. (A more rigorous justification of equation 9.5.11 is in [4].)

The method operates iteratively: For a trial value  $x$ , calculate  $a$  by equation (9.5.11). Then use  $x - a$  as the next trial value. Continue until  $a$  is sufficiently small.

The following routine implements the Laguerre method to find one root of a given polynomial of degree  $m$ , whose coefficients can be complex. As usual, the first coefficient,  $a[0]$ , is the constant term, while  $a[m]$  is the coefficient of the highest power of  $x$ . The routine implements a simplified version of an elegant stopping criterion due to Adams [5], which neatly balances the desire to achieve full machine accuracy, on the one hand, with the danger of iterating forever in the presence of roundoff error, on the other.